Graph Neural Networks With Multiple Prior Knowledge for Multi-Omics Data Analysis

Shunxin Xiao, Huibin Lin, Conghao Wang, Shiping Wang, *Member, IEEE*, and Jagath C. Rajapakse, *Fellow, IEEE*

***Abstract*—With the development of biotechnology, a large amount of multi-omics data have been collected for precision medicine. There exists multiple graph-based prior biological knowledge about omics data, such as gene-gene interaction networks. Recently, there has been an increasing interest in introducing graph neural networks (GNNs) into multi-omics learning. However, existing methods have not fully exploited these graphical priors since none have been able to integrate knowledge from multiple sources simultaneously. To solve this problem, we propose a multi-omics data analysis framework by incorporating multiple prior knowledge into graph neural network (MPK-GNN). To the best of our knowledge, this is the first attempt to introduce multiple prior graphs into multi-omics data analysis. Specifically, the proposed method contains four parts: (1) a feature-level learning module to aggregate information from prior graphs; (2) a projection module to maximize the agreement among prior networks by optimizing a contrastive loss; (3) a sample-level module to learn a global representation from input multi-omics features; (4) a task-specific module to flexibly extend MPK-GNN for various downstream multi-omics analysis tasks. Finally, we verify the effectiveness of the proposed multi-omics learning algorithm on the cancer molecular subtype classification task. Experimental results show that MPK-GNN outperforms other state-of-the-art algorithms, including multi-view learning methods and multi-omics integrative approaches.**

***Index Terms*—Multi-omics data, graph neural networks, contrastive learning, prior biological knowledge.**

Manuscript received 20 October 2022; revised 6 May 2023; accepted 2 June 2023. Date of publication 12 June 2023; date of current version 6 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grants U21A20472 and 62276065, in part by the National Key Research and Development Plan of China under Grant 2021YFB3600503, and in part by the China Scholarship Council under Grant 202106650030. *(Corresponding author: Jagath C. Rajapakse.)*

Shunxin Xiao is with the School of Computer Science and Engineering, Nanyang Technological University, Singapore 639798, and also with the College of Computer and Data Science, Fuzhou University, Fuzhou 350116, China (e-mail: xiaoshunxin.tj@gmail.com).

Huibin Lin and Shiping Wang are with the College of Computer and Data Science, Fuzhou University, Fuzhou 350116, China (e-mail: huibinlin@outlook.com; shipingwangphd@163.com).

Conghao Wang and Jagath C. Rajapakse are with the School of Computer Science and Engineering, Nanyang Technological University, Singapore 639798 (e-mail: conghao001@e.ntu.edu.sg; asjagath@ntu.edu.sg).

Digital Object Identifier 10.1109/JBHI.2023.3284794

I. Introduction

PRECISION medicine is an important aspect of future healthcare as it provides personalized treatment to patients, resulting in improved outcomes and reduced costs. For example, the same treatment may work differently for breast cancer patients due to complex clinical, pathological, and molecular characteristics [1]. With the development of high-throughput biomedical technologies, the characterizations of diseases can be formulated with multiple genomics. For example, the Cancer Genome Atlas (TCGA) dataset includes mRNA expression, DNA methylation and copy number variation (CNV) data. Compared to single-omic approaches, multi-omics methods are able to capture consistent and complementary information among multiple data to build accurate and insightful models [2]. Thus, it is necessary to introduce multi-omics data for different tasks of precision medicine, ranging from drug response prediction [3], gene discovery [4] to survival analysis [5].

Deep neural networks (DNNs) have been successfully applied to numerous areas, including natural language processing [6] and computer vision [7], due to their powerful representation capabilities. Recently, more and more researchers combined the advantages of multi-omics data and DNNs to solve various precision medicine issues [8], [9], [10]. In detail, DNNs-based approaches usually contain three stages: (a) Transforming the high-dimensional genetic features into high-level semantic embeddings; (b) Learning a unified representation from the multiple embeddings; (c) Applying the learned representation for downstream tasks. However, there are several graph structure-based prior biological knowledge, such as protein-protein interaction networks. It is difficult for traditional deep learning paradigms to utilize them effectively.

In recent years, several graph neural networks (GNNs) have been introduced to address non-Euclidean data, such as graphs and manifolds, and have demonstrated success in various fields [11], [12], [13]. For example, Wang et al. [14] proposed a graph convolutional network-based multi-omics approach that includes omics-specific learning and cross-omics correlation learning for biomedical classification. In addition, Li et al. [15] designed a GNNs-based computational model incorporated with one gene interaction network for cancer molecular subtype classification. However, these methods either do not use prior biological graph data or only use one of them. As a result, prior biological graphs were not fully utilized to improve the performance of computational models.

2168-2194 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

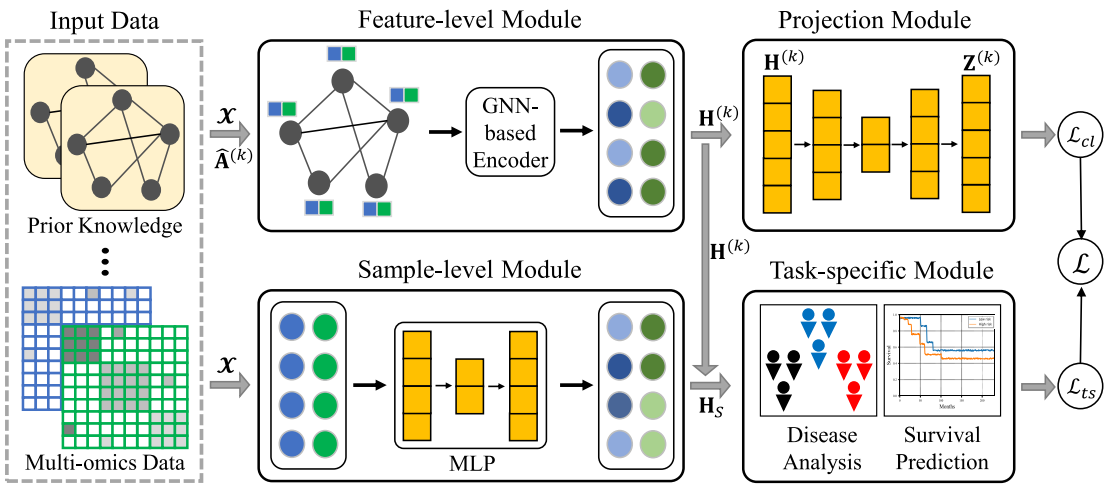

Fig. 1. Overview of the proposed multi-omics learning framework. Feature-based module aggregates information from input features to generate feature-level embeddings according to the prior graphs. Projection module is used to reconstruct a representation for each prior knowledge. In contrast, the sample-based module is implemented by a multilayer perceptron (MLP) to learn sample-level representation. Then, these multi-level representations are concatenated for the downstream tasks. Finally, the proposed MPK-GNN framework is optimized by a combined objective function $\mathcal{L}$.

To tackle the aforementioned issues, we propose a new end-to-end learning framework that incorporates multiple prior knowledge into graph neural networks (MPK-GNN) for multi-omics data analysis. The proposed framework is comprised of four modules: the feature-based learning module, the projection module, the sample-based learning module, and the task-specific module, as depicted in Fig. 1. The feature-based module is a shared graph convolutional network that takes the multi-omics features with multiple prior knowledge as inputs to learn feature-level representations. Then, the learned local embeddings are used as the inputs of a shared projection function to maximize the agreement among them by optimizing a contrastive objective. Note that the feature-based module and the projection module constitute a common contrastive learning framework. Besides, the sample-based module is designed to learn a global embedding for each input sample. Finally, the feature-level embeddings and sample-level representation are concatenated and fed into the task-specific module as input. The whole framework is optimized with a combined objective that contains the contrastive loss and the task-specific optimization loss. Specifically, the cancer molecular subtype classification task and its application in breast cancer are solved in this work. The main contributions of this work are summed up as follows:

- We propose a new extensible end-to-end deep learning framework to learn multi-level representations for various multi-omics data analysis tasks.
- To the best of our knowledge, it is the first time that a contrastive learning framework is adopted to incorporate multiple prior knowledge into GNNs simultaneously.
- Experimental results show that MPK-GNN achieves significant improvements against state-of-the-art methods on a popular TCGA dataset.

The structure of this article is organized as follows. A brief introduction to the related literature, including GNNs, multi-view learning and multi-omics analysis, is reviewed in Section II. Then, the detail of the proposed MPK-GNN framework is demonstrated in Section III. We evaluate and analyze the effectiveness of MPK-GNN in Section IV. Finally, conclusions and possible directions are drawn in Section V.

II. Related Work

A review of several related methods of multi-omics data analysis and graph neural networks is introduced in this section. We also give a brief comparison between MPK-GNN and existing methods. Finally, we introduce several popular multi-view learning methods.

A. Multi-Omics Analysis

Different from single-omic methods, multi-omics data analysis algorithms aim to integrate multiple input features for downstream tasks effectively and have been extensively explored [4], [16]. In the past few years, various deep learning paradigms were introduced for multi-omics data analysis to learn a low-dimensional representation [5], [17]. Chaudhary et al. [8] proposed a multilayer autoencoder-based framework to learn a compact embedding from three omics data for hepatocellular carcinoma survival prediction. Sharifi-Noghabi et al. [3] presented a DNNs-based multi-omics integration model, which consists of several omic-specific encoding submodules and a union representation learning module. To differentiate the status of pan-cancer metastasis, Albaradei et al. [9] designed a convolutional variational autoencoder with three heterogeneous layers based on three types of genomics data. Furthermore, Zhang et al. [10] developed a unified multi-task deep model, including a representation learning module and a task-specific module, for multi-omics data analysis tasks.

Although the aforementioned multi-omics learning methods have achieved comparable performance, they usually only use multi-omics features and are unable to exploit biological prior

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

knowledge to improve results. In contrast, the proposed GNNs-based framework is suitable to take advantage of multiple prior knowledge further to improve the robustness and performance of the computational model.

B. Graph Neural Networks

In recent years, several works have introduced graph neural networks for bioinformatics domain since there exists several graph-structured data, such as protein-protein interaction networks [18], [19]. Furthermore, some researchers are devoted to applying various graph networks for multi-omics data analysis [14], [15]. Liu et al. [20] designed a cancer drug response computational model that consists of several submodules and a unified graph convolutional network, based on the multi-omics profiles of cancer cells. Peng et al. [21] applied graph convolutional on a heterogeneous graph to learn the latent representations of both cell lines and drugs, and then built a cancer cell line-drug correlation matrix for drug response prediction. Similarly, Feng et al. [13] conducted message passing on the constructed multi-edge graphs for drug response prediction. For molecular subtype classification, Yin et al. [22] proposed a GNNs-based framework to learn a new representation for each sample by aggregating information from the related subgraphs.

The difference between MPK-GNN and the aforementioned methods is that the proposed method combines contrastive learning objective with graph convolutional network to exploit multiple prior graphs simultaneously. Consequently, these prior graphs make MPK-GNN more robust, especially when only a few supervised samples and omic features are available.

C. Multi-View Learning

Multi-view learning is a popular topic in machine learning and numerous methods have been proposed for various vision and speech tasks [23], [24]. The goal of multi-view learning methods is also to capture the consistency and complementarity information among multiple inputs. Thus, it is promising to apply them to multi-omics analysis tasks. For example, Nguyen et al. [25] have demonstrated the advantage of multi-view methods for learning the heterogeneity of samples and revealing cross-talk patterns. In this work, we also compare several multi-view learning algorithms, including both traditional machine learning and deep learning paradigms, with the proposed multi-omics integration framework.

III. Proposed Methodology

We introduce the proposed multi-omics learning framework in this section. MPK-GNN contains four submodules: the feature-level embedding learning module, the projection module, the sample-level representation learning module, and the task-specific module. Further details about these submodules and the optimization objective are elaborated below.

A. Preliminary

Multi-omics data analysis aims to learn a comprehensive embedding from multiple input features $\mathcal{X} = \{\mathbf{X}^{(v)}\}_{v=1}^{V}$ for the downstream tasks. Here, $\mathbf{X}^{(v)} \in \mathbb{R}^{n\times m}, v = 1, 2, \ldots, V$ represents the v-th omic feature. n, m and V denote the number of samples, genes and omics, respectively. Based on the input genes, several prior graphs $\mathcal{G}^{(k)} = (\mathcal{V}, \mathcal{E}^{(k)}), k = 1, 2, \ldots, K$ are generated according to the existing knowledge database, such as BioGrid [26] and STRING [27]. Here, the node set $\mathcal{V}$ represents the selected genes, $\mathcal{E}^{(k)}$ represents the edge set of the k-th prior knowledge and K stands for the total number of prior knowledge graphs. The relationships among genes are built according to certain rules. For example, the interactions among proteins with a strong normalized correlation (>0.6) will be added to the edge set. Note that the built $\mathcal{G}^{(k)}$ is different from those used in other domains since the elements of the node set $\mathcal{V}$ are genetic features instead of samples. Thus, the information aggregation mechanism is applied to the feature level.

Based on the constructed prior knowledge graph $\mathcal{G}$, the corresponding adjacency matrix $\mathbf{A} \in \mathbb{R}^{m\times m}$ can be built according to the rule $\mathbf{A}_{ij} = e_{ij}$. In detail, $e_{ij} = 1$ if there exist an edge between v_i and v_j, otherwise $e_{ij} = 0$. To reduce the effect of gradient vanishing or exploding issues appearing in deep networks, a renormalization trick form of $\mathbf{A}$ is used in this work. Concretely, the refined form is formulated as:

$$\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}. \tag{1}$$

Here, $\tilde{\mathbf{A}}$ is the self-connection form of $\mathbf{A}$ and $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij} \in \mathbb{R}^{m\times m}$ is the degree matrix of $\tilde{\mathbf{A}}$.

B. Methodology of MPK-GNN

In bioinformatics processing, there are several graph-based prior data, such as the interactions between genes. These prior graphs can offer helpful knowledge to improve the performance and maintain the robustness of the learning algorithms, especially when there is little information available. Thus, it is important to introduce these prior knowledge into multi-omics learning methods simultaneously.

Shared Architecture: Drawing inspiration from data augmentation in [28], we design a GNN-based shared contrastive learning architecture, which consists of the projection module and the feature-based module, to learn closely aligned representations by maximizing the agreement among prior graphs.

The feature-level module contains a multilayer graph convolutional encoder, a max-pooling layer and a flatten layer. Here, we denote the shared graph encoder as $f_G(\mathbf{X}_G, \hat{\mathbf{A}}^{(k)}; \Theta)$, where $\mathbf{X}_G \in \mathbb{R}^{m\times nV}$ represents the input of the feature-level module by reshaping the input multi-omics features $\mathcal{X}$. Besides, $\hat{\mathbf{A}}^{(k)}$ is the propagation matrix of the k-th prior graph obtained by (1), $\Theta = \{\mathbf{W}_G^{(l)}, \mathbf{b}_G^{(l)}\}_{l=1}^{L}$ denotes the trainable shared weights and bias of the graph encoder and L represents the total number of layers. Thus, the computation of the l-th layer of f_G on the k-th prior graph is formulated as follows:

$$\mathbf{H}_G^{(k,l)} = \sigma\left(\hat{\mathbf{A}}^{(k)}\mathbf{H}_G^{(k,l-1)}\mathbf{W}_G^{(l)} + \mathbf{b}_G^{(l)}\right). \tag{2}$$

Here, $\mathbf{H}_G^{(k,0)} = \mathbf{X}_G$ is the initial input and σ represents the used non-linear activation function. Note that the output of

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

the last layer on each prior knowledge will be reshaped as $\mathbf{H}_G^{(k)} \in \mathbb{R}^{n \times d_L \times m}$ for the next submodule, where d_L represents the dimension of the trained representations.

After the message passing stage, a graph max-pooling layer is applied for the learned embedding $\mathbf{H}_G^{(k)}$ to reduce the dimension and form a more compact representation. Here, the max-pooling layer is denoted as follows:

$$\mathbf{H}_M^{(k)} = \mathrm{MaxPooling}(\mathbf{H}_G^{(k)}; s), \tag{3}$$

where s denotes the size of the max-pooing operation and $\mathbf{H}_M^{(k)} \in \mathbb{R}^{n \times d_L \times d_M}$ is the pooled embedding with $d_M = m/s$. In the flatten layer, the output of the max-pooling layer is first flattened into a matrix $\mathbf{H}_F^{(k,0)} \in \mathbb{R}^{n \times d_L d_M}$, which serves as the input to a shared non-linear transformation network that learns the final output $\mathbf{H}^{(k)} \in \mathbb{R}^{n \times d_F}$ of the feature-level module.

The projection module is designed to reconstruct a representation for each prior knowledge. We implement this module by a shallow neural network and denote it as $f_P(\mathbf{H}^{(k)}; \Omega)$, where $\Omega = \{\mathbf{W}_P^{(t)}, \mathbf{b}_P^{(t)}\}_{t=1}^{T}$ and T are the learnable shared parameters and the number of layers, respectively. The output of the t-th layer for the k-th prior graph is computed as:

$$\mathbf{Z}^{(k,t)} = \sigma\left(\mathbf{Z}^{(k,t-1)}\mathbf{W}_P^{(t)} + \mathbf{b}_P^{(t)}\right), \tag{4}$$

where $\mathbf{Z}^{(k,0)} = \mathbf{H}^{(k)}$ is the initial input and $\mathbf{Z}^{(k)} = \mathbf{Z}^{(k,T)}$ stands for the projection representation of the k-th input channel. The agreement among $\{\mathbf{Z}^{(k)}\}_{k=1}^{K}$ can be maximized by optimizing a contrastive learning objective.

Sample-based Module: We introduce a sample-based parallel module to learn a global representation for each sample. This module is implemented by a multilayer perceptron $f_S(\mathbf{X}_S; \Psi)$ that the computation process is shown as follows:

$$\mathbf{H}_S^{(i)} = \sigma\left(\mathbf{H}_S^{(i-1)}\mathbf{W}_S^{(i)} + \mathbf{b}_S^{(i)}\right), \tag{5}$$

where $\mathbf{H}_S^{(0)} = \mathbf{X}_S \in \mathbb{R}^{n \times mV}$, $\mathbf{H}_S^{(i)}$ and $\mathbf{H}_S = \mathbf{H}_S^{(I)} \in \mathbb{R}^{n \times d_S}$ represent the original input, the output of the i-th layer and the final learned global representation, respectively. Besides, $\Psi = \{\mathbf{W}_S^{(i)}, \mathbf{b}_S^{(i)}\}_{i=1}^{I}$ denotes the trainable weights and bias, I is the number of layers of f_S, and d_S is the output dimension. The function σ only be applied to the first $I-1$ layers.

Task-specific Module: Based on the abovementioned modules, we design a task-specific module, which can be implemented by different networks according to the target tasks. The learned global feature $\mathbf{H}_S$ and all local features $\{\mathbf{H}^{(k)}\}_{k=1}^{K}$ are concatenated to form a unified input for this module:

$$\mathbf{U} = [\mathbf{H}_S, \mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(K)}]. \tag{6}$$

Here, each sample is represented as a $(d_S + d_F K)$-dimension vector. Taking the cancer molecular subtype classification as an example, we feed $\mathbf{U}$ into a fully-connected network followed by a $\mathrm{softmax}$ function to obtain the probability distribution matrix of labels.

C. Training of MPK-GNN

MPK-GNN is optimized by a combined objective consisting of contrastive learning loss $\mathcal{L}_{cl}$ and task-specific learning objective $\mathcal{L}_{ts}$. For $\mathcal{L}_{cl}$, we stack all reconstructed representations $\{\mathbf{Z}^{(k)}\}_{k=1}^{K}$ to form a set of node representations $\mathbf{Z}$ with nK rows and use $f \in \mathcal{F} \equiv \{1, \ldots, nK\}$ to denote the index of each node. Then, the contrastive loss $\mathcal{L}_{cl}$ is formed as follows:

$$\mathcal{L}_{cl} = \sum_{f \in \mathcal{F}} \frac{-1}{|\mathcal{P}(f)|} \sum_{p \in \mathcal{P}(f)} \log \frac{\exp\left(\mathbf{Z}_f \cdot \mathbf{Z}_p/\beta\right)}{\sum_{a \in \mathcal{A}(f)} \exp\left(\mathbf{Z}_f \cdot \mathbf{Z}_a/\beta\right)}. \tag{7}$$

Here, $\mathbf{Z}_f$ represents the f-th row of $\mathbf{Z}$, β is a scale hyper-parameter, and $\mathcal{P}(f)$ is the set of indices where the indicated samples and the anchor node v_f have the same label. Besides, $\mathcal{A}(f) \equiv \mathcal{F} \backslash \{f\}$ represents the set of indices of all samples except the target node v_f, and $\cdot$ represents the inner (dot) product. For unsupervised learning tasks, $\mathcal{P}(f)$ will only contain the indices where the indicated rows and f-th row stand for the different views of the same original sample.

For various application scenarios, the task-specific objective $\mathcal{L}_{ts}$ is implemented by different optimization functions. For example, we utilize the popular cross-entropy as the optimization objective when applying MPK-GNN for classification tasks and the form of $\mathcal{L}_{ts}$ is shown as follows:

$$\mathcal{L}_{ts} = -\frac{1}{Y} \sum_{y=1}^{Y} \sum_{c=1}^{C} \mathcal{I}_{yc} \log\left(p_{yc}\right). \tag{8}$$

Here, Y represents the number of training samples, C stands for the number of classes and p_{yc} is the prediction probability of node v_y belonging to the c-th class. Besides, $\mathcal{I}$ is an indicator function and $\mathcal{I}_{yc} = 1$ if the label of v_y is c. Finally, the combined optimization loss of the proposed multi-omics learning framework can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{ts} + \lambda \mathcal{L}_{cl}, \tag{9}$$

where λ is a coefficient to adjust the importance of the contrastive learning objective. The weights of MPK-GNN can be updated by the back-propagation process. Consequently, the learned representation is more suitable for the target task by optimizing the combined objective.

IV. Experiments

In this section, the effectiveness of the proposed MPK-GNN framework is demonstrated via the cancer molecular subtype classification task and its application in breast cancer. We first introduce the experimental setup, and then the classification results of all algorithms are shown to demonstrate the superiority of MPK-GNN. Finally, we conduct several additional experiments to analyze the sensitivity of different hyper-parameters or implementations.

A. Experimental Setup

Datasets: The popular TCGA Pan-cancer dataset [29] with two omics, including gene expression and CNV data, is utilized to solve the cancer molecular subtype classification task. We

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

closely follow the pre-processing steps used in Li et al. [15]. The gene expression data includes 11,060 samples and the CNV data consists of 10,845 samples. After filtering for missing data, the final used dataset contained a total of 9,664 samples. All samples consist of 17,944 genetic features and belonged to 28 classes. Then, the top m genetic features were selected according to the variances among them after normalizing the used multi-omics data. Concretely, the number of features m varies in $\{100, 200, 500, 1{,}000, 2{,}000, 5{,}000\}$ and the corresponding dataset is denoted as TCGA-m. For breast cancer subtype recognition, the breast invasive carcinoma (BRCA) dataset consists of 606 patient samples with two omics (mRNA and CNV) utilized in the experiments. BRCA contains five subtypes of breast cancer, which are luminal A, luminal B, HER2(+), TNBC and unclear. Both mRNA and CNA are represented by 58 genetic features.

Three prior biological knowledge, including gene-gene interaction (GGI) networks, protein-protein interaction (PPI) networks, and co-expression (CoExp) networks, were utilized in the experiments. The GGI graph was derived from the BioGrid database [26] and both the PPI graph and CoExp graph were constructed from the STRING database [27].

Compared Methods: To validate the effectiveness of the proposed MPK-GNN framework, we compared the performance of two classification tasks with different types of approaches, ranging from the traditional machine learning methods to the state-of-the-art deep learning models. In detail, SVM, RF and KNN are classical single-view learning methods. Conversely, AMGL, MVAR, MLAN, Co-GCN and ERL-MVSC belong to multi-view learning algorithms and were proposed for image or textual analysis. Finally, DeepMO, MOGONET and CMSC are recently proposed multi-omics data analysis approaches.

Implementation Details: For the single-view baselines, the input multi-omics data will be concatenated to form a unified feature matrix. To have a fair comparison, we performed hyper-parameter tuning for the used compared methods except for the parameter-free method AMGL. For example, the number of decision trees used in RF was tuned to be 20 and the number of the nearest neighbors was decided as 1 for KNN. Note that the tuning experiments are conducted on TCGA-1000 with 10% training samples and then the tuned values are applied to the whole datasets with various labeling ratios.

For the proposed method, the hidden dimensions of the sample-based learning module are set constant to 1,024 and 256 for all datasets. In the feature-level learning module, the dimensional distribution of the graph convolutional encoder is 2,048 and 512 and the size of the max-pooling layer is set to 8. The hidden dimension of the flatten layer is set to 32 and the shared projection function has two layers with the size of 2048 and 32. For the training process, the momentum of the SGD optimizer is set to 0.9 and the mini-batch size is tuned as 128. Similar to the setting of compared methods, the aforementioned hyper-parameters of MPK-GNN are tuned on TCGA-1000 with 10% training samples and then used for other situations. We utilized the popular deep learning package PyTorch with the 1.8 version to implement the proposed method. For all algorithms, we repeat each experiment ten times and use the mean with standard deviation results as the final performance. All experiments are conducted on an Ubuntu-16.04 system, which contains a 2.1 GHz Intel Xeon CPU with 128 GB RAM and an Nvidia Tesla P100 GPU with 16 GB memory. The implementation of MPK-GNN is available at https://github.com/Xiaoshunxin/MPK-GNN.

B. Cancer Molecular Subtype Classification

Comprehensive experiments were conducted to validate the performance of the proposed MPK-GNN method in the cancer molecular subtype classification. In Tables I and II, the classification results of all compared algorithms on six datasets are reported under different percentages of training samples. For intuitive comparison, the best results were highlighted in boldface. MPK-GNN achieves the best results on the used datasets under different labeling rates. It demonstrates that MPK-GNN is able to achieve significant performance in both semi-supervised and supervised tasks.

Furthermore, several interesting phenomena can be observed by analyzing the classification results. Firstly, the proposed multi-omics learning framework outperforms all compared single-view methods. However, not all multi-view or multi-omics methods achieve better performance than the single-view baselines over various labeling rates. It indicates that the proper utilization of the input multiple features is crucial to multi-omics data analysis. Compared to other deep learning-based methods, MPK-GNN achieves the smallest average standard deviations under running multiple times. In this respect, the proposed framework is more robust against these methods. Note that the robustness of a deep learning framework is important for the applications of bioinformatics processing. Furthermore, CMSC achieves poor performance on TCGA-100 and TCGA-200 under low labeling rates, such as 10% and 20%. One possible reason is that there is not enough supervised information to train the learning model. Specifically, the number of training samples and the number of gene features are both small in the abovementioned setting. However, the proposed MPK-GNN achieves remarkable performance in the same situation by introducing more prior information. This indicates that the utilization of multiple prior knowledge is critical for multi-omics data analysis, especially in the case of semi-supervised learning. As another GNNs-based method, the performance of MOGONET is worse than CMSC and the proposed MPK-GNN in all situations. A possible reason is that the adjacency matrices used for message passing in MOGONET are constructed by the cosine similarity between pairs of nodes. However, these simulated adjacency graphs are easily affected by the quality of the input feature space, which may cause incorrect information propagation. Another possible reason is that MOGONET fails to utilize the existing prior information to guide the learning process. This further indicates the superiority of the utilization of multiple prior knowledge. The benefits of using multiple prior knowledge graphs are not significant compared to using one graph when the number of genetic features or training samples is large. One possible explanation is that as the number of training samples or genes increases, the model receives more informative data, reducing the effectiveness of using multiple prior graphs.

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

TABLE I
THE ACCURACY (MEAN% ± STD%) COMPARISON OF ALL COMPARED APPROACHES ON THREE CANCER MOLECULAR SUBTYPE CLASSIFICATION DATASETS. THE BEST RESULTS ARE MARKED IN BOLD (THE HIGHER THE BETTER)

Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-100	SVM	72.0 (0.0)	74.0 (0.0)	76.1 (0.0)	76.9 (0.0)	77.6 (0.0)	78.8 (0.0)	78.3 (0.0)	80.4 (0.0)
	RF	72.1 (0.4)	73.8 (0.4)	75.0 (0.2)	75.7 (0.5)	76.3 (0.3)	77.0 (0.4)	77.1 (0.6)	78.4 (0.5)
	KNN	65.5 (0.0)	67.2 (0.0)	68.2 (0.0)	68.3 (0.0)	68.7 (0.0)	69.0 (0.0)	68.6 (0.0)	69.6 (0.0)
	AMGL	65.3 (0.4)	68.7 (0.3)	71.3 (0.3)	72.7 (0.3)	73.5 (0.6)	74.4 (0.4)	74.5 (0.5)	75.4 (0.8)
	MVAR	71.3 (0.7)	72.0 (0.5)	72.1 (0.5)	72.1 (0.6)	72.2 (0.3)	72.4 (0.5)	72.5 (0.5)	72.5 (0.6)
	MLAN	70.8 (0.6)	72.5 (0.5)	72.8 (0.5)	72.4 (0.9)	73.5 (0.8)	72.7 (1.4)	72.8 (0.7)	71.6 (2.5)
	Co-GCN	65.7 (0.8)	69.6 (0.6)	70.7 (0.3)	70.8 (0.3)	71.1 (0.4)	73.2 (0.3)	73.5 (0.3)	71.9 (0.5)
	DeepMo	72.0 (0.4)	73.9 (0.5)	74.4 (0.2)	75.1 (0.2)	75.8 (0.2)	76.2 (0.2)	76.1 (0.3)	76.6 (0.5)
	MOGONET	42.9 (1.9)	50.7 (1.8)	49.6 (1.6)	57.3 (1.2)	68.2 (0.5)	69.5 (0.8)	68.5 (0.8)	65.8 (1.0)
	ERL-MVSC	63.7 (0.6)	71.0 (0.3)	73.8 (0.4)	75.2 (0.3)	75.8 (0.5)	76.5 (0.6)	77.0 (0.7)	76.6 (0.9)
	CMSC	63.9 (1.2)	73.3 (0.5)	75.5 (0.4)	76.7 (0.3)	77.6 (0.2)	78.7 (0.2)	78.2 (0.4)	79.9 (0.5)
	MPK-GNN	**72.9 (0.6)**	**75.0 (0.2)**	**77.1 (0.3)**	**77.8 (0.2)**	**78.6 (0.2)**	**79.9 (0.1)**	**79.9 (0.5)**	**81.2 (0.3)**
Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-200	SVM	72.9 (0.0)	75.1 (0.0)	76.8 (0.0)	77.8 (0.0)	78.4 (0.0)	79.1 (0.0)	78.8 (0.0)	80.3 (0.0)
	RF	72.5 (0.5)	74.5 (0.1)	75.5 (0.2)	76.3 (0.3)	76.9 (0.4)	77.5 (0.6)	77.3 (0.5)	78.5 (0.9)
	KNN	66.0 (0.0)	68.4 (0.0)	69.2 (0.0)	69.3 (0.0)	70.4 (0.0)	70.5 (0.0)	69.5 (0.0)	70.8 (0.0)
	AMGL	66.6 (0.4)	70.2 (0.3)	72.3 (0.3)	73.7 (0.3)	74.7 (0.5)	75.2 (0.5)	76.0 (0.7)	75.8 (0.9)
	MVAR	73.3 (0.5)	73.5 (0.6)	73.5 (0.4)	73.5 (0.4)	73.9 (0.7)	73.7 (0.7)	73.4 (0.6)	73.5 (0.8)
	MLAN	71.4 (0.8)	73.5 (0.4)	74.7 (0.4)	74.9 (0.5)	75.1 (0.8)	75.5 (0.4)	75.6 (0.8)	75.7 (1.0)
	Co-GCN	67.6 (1.0)	70.8 (0.6)	72.1 (0.5)	73.3 (0.2)	73.6 (0.4)	75.6 (0.3)	76.1 (0.3)	74.0 (0.3)
	DeepMo	73.2 (0.4)	74.8 (0.2)	75.1 (0.3)	75.5 (0.2)	76.4 (0.3)	77.0 (0.4)	76.7 (0.3)	77.7 (0.4)
	MOGONET	42.9 (1.9)	50.7 (1.8)	49.6 (1.6)	57.3 (1.7)	68.2 (0.5)	69.5 (0.8)	68.5 (0.8)	65.8 (1.0)
	ERL-MVSC	65.1 (1.0)	72.4 (0.3)	74.9 (0.3)	76.2 (0.3)	76.8 (0.5)	77.1 (0.5)	77.7 (0.7)	77.8 (0.8)
	CMSC	70.2 (0.5)	75.6 (0.6)	77.4 (0.2)	78.3 (0.2)	78.8 (0.4)	79.5 (0.3)	79.5 (0.4)	80.7 (0.7)
	MPK-GNN	**74.3 (0.4)**	**77.3 (0.3)**	**78.9 (0.3)**	**79.4 (0.2)**	**79.9 (0.2)**	**81.2 (0.4)**	**81.1 (0.3)**	**82.4 (0.4)**
Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-500	SVM	73.8 (0.0)	76.7 (0.0)	78.6 (0.0)	79.4 (0.0)	80.4 (0.0)	81.3 (0.0)	80.7 (0.0)	82.3 (0.0)
	RF	74.0 (0.3)	75.9 (0.4)	76.7 (0.3)	77.7 (0.4)	78.1 (0.5)	78.4 (0.4)	78.6 (0.6)	79.7 (0.5)
	KNN	67.0 (0.0)	69.9 (0.0)	70.0 (0.0)	70.9 (0.0)	71.6 (0.0)	72.5 (0.0)	71.4 (0.0)	72.3 (0.0)
	AMGL	67.6 (0.6)	71.6 (0.3)	73.5 (0.3)	75.0 (0.3)	75.9 (0.4)	76.7 (0.5)	77.5 (0.4)	77.5 (0.8)
	MVAR	75.7 (0.4)	76.5 (0.3)	76.8 (0.6)	76.5 (0.7)	77.1 (0.5)	77.0 (0.6)	77.2 (1.1)	77.4 (0.8)
	MLAN	74.3 (0.4)	75.6 (0.4)	76.5 (0.6)	77.2 (0.4)	77.4 (0.6)	77.7 (0.3)	77.9 (0.6)	77.7 (0.8)
	Co-GCN	69.9 (0.5)	71.9 (0.6)	73.9 (0.6)	75.0 (0.3)	75.9 (0.2)	77.9 (0.2)	78.1 (0.3)	76.7 (0.3)
	DeepMo	75.7 (0.4)	77.0 (0.4)	77.5 (0.3)	78.7 (1.0)	78.6 (0.3)	79.2 (0.2)	79.2 (0.4)	79.8 (0.3)
	MOGONET	45.4 (1.3)	55.2 (1.2)	55.8 (1.4)	61.8 (1.0)	71.1 (1.6)	71.6 (1.6)	71.2 (1.4)	69.3 (2.1)
	ERL-MVSC	67.4 (0.5)	74.3 (0.6)	76.8 (0.4)	77.9 (0.4)	78.2 (0.4)	78.9 (0.5)	78.7 (0.8)	79.2 (0.9)
	CMSC	75.2 (0.4)	78.8 (0.2)	80.3 (0.2)	81.0 (0.3)	81.5 (0.2)	82.0 (0.4)	82.0 (0.5)	83.9 (0.4)
	MPK-GNN	**77.5 (0.5)**	**80.1 (0.4)**	**81.4 (0.3)**	**82.1 (0.3)**	**82.5 (0.3)**	**83.2 (0.3)**	**83.0 (0.3)**	**85.2 (0.5)**

C. Application in Breast Cancer

To verify the superiority of the proposed multi-omics data analysis model, we further compared the performance of various learning methods on the breast cancer subtype classification task. In Table III, we report the classification accuracies of eleven algorithms on the BRCA dataset with different ratios of labeled data. Experimental results show that the proposed MPK-GNN framework outperforms the compared methods under different labeling rates. The observation demonstrates that MPK-GNN is also effective in specific applications, such as breast cancer analysis. In addition, CMSC achieved better performance than other compared methods in most cases. One possible reason is that CMSC also takes the prior knowledge graph as input. It proves once again the importance of using prior knowledge networks, especially when the number of genetic features or training samples is small.

D. Model Analysis

In this section, several additional experiments are designed to validate the proposed multi-omics learning framework. Note that most of the experimental settings are the same as the statements in Section IV-A except for the factors considered for the current analysis.

Effect of Multiple Prior Knowledge: We conduct an additional study to validate the effectiveness of using multiple prior knowledge simultaneously. Specifically, the CMSC method is utilized to obtain the baseline performance where only one prior knowledge is used in the learning framework. Here, three prior knowledge, including GGI, PPI and CoExp, are used as the input of CMSC alone. Furthermore, we replace the prior graph-based module of CMSC with a commonly used GCN to release the requirement of prior knowledge. We denote the baseline as Basic, in which the required adjacency matrix is constructed based on the input feature space. As shown in Table IV, MPK-GNN outperforms all single prior knowledge variants and the baseline on all benchmark datasets under different labeling ratios. Compared to single variants, the proposed multiple prior knowledge method achieves a significant improvement when the information on gene features and supervised samples are both scarce. By utilizing CoExp and GGI networks as input graphs, models tend to exhibit better prediction accuracy compared to those employing the PPI network. This phenomenon can be attributed to the fact that the interactions within the PPI graph

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

TABLE II

THE ACCURACY (MEAN% ± STD%) COMPARISON OF ALL COMPARED APPROACHES ON THREE CANCER MOLECULAR SUBTYPE CLASSIFICATION DATASETS. THE BEST RESULTS ARE MARKED IN BOLD (THE HIGHER THE BETTER)

Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-1000	SVM	72.7 (0.0)	76.2 (0.0)	79.0 (0.0)	79.7 (0.0)	80.4 (0.0)	81.1 (0.0)	80.4 (0.0)	82.3 (0.0)
	RF	73.9 (0.4)	76.0 (0.3)	77.0 (0.4)	77.8 (0.4)	78.4 (0.4)	79.0 (0.5)	78.9 (0.6)	80.0 (0.8)
	KNN	65.2 (0.0)	68.3 (0.0)	68.9 (0.0)	69.9 (0.0)	70.4 (0.0)	70.5 (0.0)	69.5 (0.0)	70.7 (0.0)
	AMGL	67.7 (0.4)	71.6 (0.2)	73.7 (0.4)	75.0 (0.7)	76.2 (0.3)	76.9 (0.6)	77.3 (0.7)	77.8 (0.5)
	MVAR	76.0 (0.4)	76.9 (0.4)	77.9 (0.4)	77.4 (0.5)	77.9 (0.3)	77.6 (0.9)	78.0 (0.7)	78.1 (0.8)
	MLAN	73.7 (0.4)	75.3 (0.4)	76.1 (0.4)	76.7 (0.4)	77.3 (0.6)	77.6 (0.5)	78.0 (0.7)	77.9 (1.3)
	Co-GCN	70.9 (0.3)	72.6 (0.4)	74.5 (0.4)	75.4 (0.4)	75.7 (0.5)	77.8 (0.3)	77.9 (0.3)	77.3 (0.3)
	DeepMo	76.5 (0.3)	77.5 (0.3)	78.3 (0.3)	78.8 (0.4)	79.8 (1.4)	79.8 (0.4)	79.7 (0.3)	80.1 (1.6)
	MOGONET	45.5 (1.2)	54.0 (1.2)	54.7 (1.2)	61.0 (1.4)	71.4 (1.3)	72.1 (1.7)	71.1 (1.9)	69.3 (3.3)
	ERL-MVSC	67.1 (0.6)	74.3 (0.5)	76.9 (0.3)	77.9 (0.4)	78.6 (0.5)	78.9 (0.4)	79.3 (0.9)	79.8 (1.0)
	CMSC	76.0 (0.4)	79.7 (0.4)	81.2 (0.2)	82.0 (0.3)	82.7 (0.2)	82.9 (0.3)	82.9 (0.4)	84.2 (0.7)
	MPK-GNN	**78.0 (0.3)**	**80.3 (0.4)**	**81.9 (0.4)**	**83.0 (0.2)**	**83.6 (0.2)**	**84.1 (0.4)**	**84.0 (0.4)**	**85.3 (0.6)**
Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-2000	SVM	71.2 (0.0)	75.3 (0.0)	78.4 (0.0)	79.6 (0.0)	80.2 (0.0)	80.7 (0.0)	80.6 (0.0)	82.2 (0.0)
	RF	74.7 (0.3)	76.6 (0.2)	77.3 (0.4)	78.1 (0.5)	78.8 (0.5)	79.6 (0.3)	79.2 (0.5)	80.7 (0.5)
	KNN	63.4 (0.0)	66.8 (0.0)	67.4 (0.0)	68.2 (0.0)	68.6 (0.0)	68.9 (0.0)	68.1 (0.0)	69.5 (0.0)
	AMGL	67.5 (0.5)	71.7 (0.5)	73.8 (0.4)	75.1 (0.4)	76.2 (0.3)	76.8 (0.6)	77.5 (0.6)	78.1 (1.2)
	MVAR	75.9 (0.8)	77.0 (0.4)	77.5 (0.2)	78.1 (0.3)	78.0 (0.5)	78.4 (0.4)	78.4 (0.6)	78.7 (1.0)
	MLAN	73.4 (0.4)	75.1 (0.3)	75.9 (0.4)	76.6 (0.5)	76.9 (0.5)	77.4 (0.6)	77.5 (0.7)	77.5 (0.7)
	Co-GCN	70.9 (0.4)	73.0 (0.4)	74.7 (0.4)	75.6 (0.3)	75.8 (0.5)	78.0 (0.5)	77.9 (0.4)	76.7 (0.5)
	DeepMo	75.1 (2.0)	77.8 (0.5)	78.6 (0.5)	79.5 (0.2)	79.8 (0.4)	80.2 (0.3)	80.2 (0.3)	80.5 (0.4)
	MOGONET	45.4 (0.9)	54.3 (1.3)	53.5 (0.9)	61.2 (1.2)	71.5 (1.0)	73.0 (1.1)	71.0 (1.9)	70.2 (1.7)
	ERL-MVSC	67.4 (0.6)	74.9 (0.3)	77.3 (0.5)	77.9 (0.3)	78.6 (0.4)	79.0 (0.3)	79.1 (0.8)	79.8 (0.9)
	CMSC	76.8 (0.3)	80.2 (0.2)	81.9 (0.2)	83.0 (0.3)	83.5 (0.3)	84.3 (0.4)	84.2 (0.4)	85.7 (0.5)
	MPK-GNN	**77.8 (0.7)**	**81.1 (0.2)**	**82.7 (0.3)**	**83.7 (0.2)**	**84.5 (0.3)**	**84.8 (1.3)**	**85.0 (0.3)**	**86.0 (0.4)**
Dataset	Methods	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-5000	SVM	71.7 (0.0)	76.2 (0.0)	79.3 (0.0)	80.5 (0.0)	81.0 (0.0)	81.8 (0.0)	81.9 (0.0)	83.0 (0.0)
	RF	75.4 (0.4)	77.2 (0.2)	78.4 (0.5)	79.5 (0.6)	80.1 (0.4)	80.8 (0.5)	81.0 (0.5)	82.1 (0.4)
	KNN	62.2 (0.0)	66.2 (0.0)	67.2 (0.0)	68.2 (0.0)	68.8 (0.0)	69.8 (0.0)	68.9 (0.0)	70.4 (0.0)
	AMGL	68.2 (0.4)	72.0 (0.4)	74.3 (0.4)	75.6 (0.4)	76.8 (0.3)	77.4 (0.5)	78.2 (0.5)	78.8 (0.9)
	MVAR	77.7 (0.8)	79.2 (0.4)	79.9 (0.4)	80.2 (0.4)	80.4 (0.5)	80.8 (0.3)	81.0 (0.6)	80.5 (0.8)
	MLAN	73.1 (0.5)	75.0 (0.5)	76.0 (0.6)	77.1 (0.5)	77.2 (0.5)	77.8 (0.6)	77.6 (0.7)	77.9 (0.7)
	Co-GCN	72.5 (0.4)	74.8 (0.2)	76.4 (0.3)	77.2 (0.3)	77.7 (0.4)	80.2 (0.3)	79.8 (0.4)	78.4 (0.4)
	DeepMo	73.9 (2.6)	78.2 (0.8)	79.1 (0.4)	80.1 (0.4)	80.5 (0.6)	80.6 (0.5)	80.7 (0.6)	80.6 (0.4)
	MOGONET	44.9 (1.6)	54.8 (1.1)	56.9 (0.4)	62.5 (0.9)	72.8 (1.1)	72.4 (2.3)	72.2 (1.6)	69.8 (2.5)
	ERL-MVSC	68.5 (0.7)	75.5 (0.4)	78.0 (0.4)	79.1 (0.4)	79.7 (0.4)	80.4 (0.5)	80.6 (0.6)	80.6 (1.4)
	CMSC	77.9 (0.4)	81.4 (0.3)	82.8 (0.5)	84.1 (0.2)	84.9 (0.4)	85.3 (0.3)	85.1 (0.5)	86.1 (0.5)
	MPK-GNN	**78.6 (0.6)**	**81.5 (0.4)**	**83.5 (0.3)**	**84.6 (0.2)**	**85.6 (0.4)**	**85.9 (0.2)**	**85.7 (0.4)**	**86.6 (0.4)**

TABLE III

THE ACCURACY PERFORMANCE (MEAN% ± STD%) COMPARISON ON THE BREAST CANCER SUBTYPE CLASSIFICATION TASK. THE BEST RESULTS ARE MARKED IN BOLD (THE HIGHER THE BETTER)

Methods	10%	20%	30%	40%	50%	60%	70%	80%
SVM	46.2 (0.0)	47.3 (0.0)	47.3 (0.0)	48.2 (0.0)	50.6 (0.0)	51.1 (0.0)	53.3 (0.0)	57.4 (0.0)
RF	52.4 (1.8)	60.1 (2.9)	62.5 (1.1)	61.8 (2.2)	60.6 (1.9)	62.6 (2.9)	63.7 (2.3)	64.8 (3.2)
KNN	53.4 (0.0)	54.4 (0.0)	54.1 (0.0)	54.5 (0.0)	55.1 (0.0)	52.8 (0.0)	53.3 (0.0)	57.4 (0.0)
AMGL	51.5 (2.2)	54.6 (2.6)	57.0 (1.9)	57.5 (1.1)	59.1 (2.1)	59.6 (1.9)	60.4 (3.1)	61.8 (3.9)
MVAR	54.4 (3.6)	57.1 (2.3)	58.5 (3.7)	59.8 (1.4)	62.7 (1.6)	60.7 (3.3)	64.1 (4.1)	62.6 (4.5)
MLAN	47.7 (2.0)	48.4 (2.6)	49.3 (2.2)	46.3 (3.1)	45.7 (2.1)	45.1 (4.4)	43.8 (3.0)	44.9 (3.2)
Co-GCN	45.2 (1.6)	45.9 (1.2)	46.2 (2.2)	45.0 (2.4)	46.4 (1.4)	48.1 (1.4)	45.9 (3.1)	49.5 (4.6)
DeepMo	49.4 (2.1)	51.6 (0.8)	52.3 (1.2)	54.9 (1.5)	53.1 (1.5)	55.7 (2.0)	59.3 (1.6)	62.0 (2.8)
ERL-MVSC	50.6 (1.9)	55.0 (2.3)	57.7 (1.3)	58.7 (1.8)	60.1 (1.4)	60.4 (3.1)	61.7 (3.5)	62.3 (2.9)
CMSC	59.7 (2.2)	63.7 (1.2)	64.1 (1.8)	64.9 (1.3)	63.5 (1.2)	63.8 (1.0)	63.9 (1.8)	68.2 (3.5)
MPK-GNN	**62.1 (1.1)**	**63.9 (1.1)**	**64.2 (0.8)**	**65.5 (0.5)**	**64.7 (1.0)**	**64.2 (1.0)**	**66.2 (1.0)**	**69.3 (1.1)**

solely involve genes that are associated with proteins and possess established interactions, while non-coding genes lacking protein association and interactions are disregarded. Besides, we can observe that the proposed method achieves better performance than the baseline in all cases, while using a single prior knowledge does not achieve the same results. This observation demonstrates the importance of utilizing multiple prior knowledge.

Influence of Contrastive Learning: To verify the effectiveness of the shared contrastive learning framework, we conduct a study to compare the performance of the proposed

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
IMPACT OF MULTIPLE PRIOR KNOWLEDGE MEASURED BY CLASSIFICATION ACCURACY (MEAN% ± STD%) ON THE CANCER MOLECULAR SUBTYPE CLASSIFICATION TASK. THE BEST RESULTS ARE MARKED IN BOLD (THE HIGHER THE BETTER)

Dataset	Types	10%	20%	30%	40%	50%	60%	70%	80%
TCGA-100	Basic	71.9 (0.4)	74.0 (0.1)	75.1 (0.2)	75.5 (0.3)	76.0 (0.3)	76.7 (0.4)	76.1 (0.5)	76.7 (0.8)
	GGI	63.9 (1.2)	73.3 (0.5)	75.5 (0.4)	76.7 (0.3)	77.6 (0.2)	78.7 (0.2)	78.2 (0.4)	79.9 (0.5)
	PPI	63.9 (1.1)	73.2 (0.4)	75.2 (0.4)	76.6 (0.3)	77.5 (0.3)	78.6 (0.2)	78.1 (0.4)	79.7 (0.5)
	CoExp	63.8 (1.2)	73.4 (0.3)	75.5 (0.4)	76.7 (0.3)	77.6 (0.2)	78.6 (0.3)	78.2 (0.3)	79.8 (0.3)
	All	**72.9 (0.6)**	**75.0 (0.2)**	**77.1 (0.3)**	**77.8 (0.2)**	**78.6 (0.2)**	**79.9 (0.1)**	**79.9 (0.5)**	**81.2 (0.3)**
TCGA-200	Basic	73.1 (0.5)	75.4 (0.4)	76.6 (0.3)	77.2 (0.4)	77.6 (0.2)	78.1 (0.4)	77.5 (0.5)	78.8 (0.6)
	GGI	70.2 (0.5)	75.6 (0.6)	77.4 (0.2)	78.3 (0.2)	78.8 (0.4)	79.5 (0.3)	79.5 (0.4)	80.7 (0.7)
	PPI	70.3 (0.5)	75.6 (0.3)	77.4 (0.3)	78.3 (0.4)	78.8 (0.4)	79.5 (0.3)	79.4 (0.3)	80.4 (0.5)
	CoExp	70.2 (0.4)	75.7 (0.5)	77.5 (0.3)	78.4 (0.3)	78.8 (0.4)	79.6 (0.4)	79.4 (0.3)	80.4 (0.6)
	All	**74.3 (0.4)**	**77.3 (0.3)**	**78.9 (0.3)**	**79.4 (0.2)**	**79.9 (0.2)**	**81.2 (0.4)**	**81.1 (0.3)**	**82.4 (0.4)**
TCGA-500	Basic	75.3 (0.5)	77.8 (0.4)	79.0 (0.5)	79.5 (0.6)	80.1 (0.6)	80.5 (0.6)	79.8 (0.7)	81.9 (0.9)
	GGI	75.2 (0.4)	78.8 (0.2)	80.2 (0.2)	81.0 (0.3)	81.5 (0.2)	82.0 (0.4)	82.0 (0.5)	83.9 (0.4)
	PPI	75.0 (0.8)	78.7 (0.3)	80.2 (0.2)	80.8 (0.3)	81.5 (0.3)	82.1 (0.3)	81.8 (0.5)	84.0 (0.4)
	CoExp	75.1 (0.5)	78.8 (0.3)	80.3 (0.2)	81.0 (0.2)	81.6 (0.3)	82.0 (0.4)	82.1 (0.3)	84.1 (0.4)
	All	**77.5 (0.5)**	**80.1 (0.4)**	**81.4 (0.3)**	**82.1 (0.3)**	**82.5 (0.3)**	**83.2 (0.3)**	**83.0 (0.3)**	**85.2 (0.5)**
TCGA-1000	Basic	75.1 (0.3)	77.9 (0.4)	79.5 (0.3)	80.2 (0.4)	81.0 (0.4)	81.4 (0.3)	81.3 (0.5)	83.0 (0.8)
	GGI	76.0 (0.4)	79.7 (0.4)	81.2 (0.2)	82.0 (0.3)	82.7 (0.2)	82.9 (0.3)	82.9 (0.4)	84.2 (0.7)
	PPI	75.9 (0.5)	79.6 (0.4)	81.1 (0.2)	81.7 (0.4)	82.6 (0.3)	82.9 (0.3)	82.7 (0.4)	84.3 (0.6)
	CoExp	76.1 (0.5)	79.8 (0.4)	81.1 (0.3)	81.9 (0.4)	82.6 (0.2)	83.0 (0.3)	82.9 (0.4)	84.2 (0.6)
	All	**78.0 (0.3)**	**80.3 (0.4)**	**81.9 (0.4)**	**83.0 (0.2)**	**83.6 (0.2)**	**84.1 (0.4)**	**84.0 (0.4)**	**85.3 (0.6)**
TCGA-2000	Basic	73.9 (0.5)	77.4 (0.6)	79.4 (0.5)	80.3 (0.3)	80.9 (0.5)	81.3 (0.5)	82.0 (0.5)	82.8 (0.8)
	GGI	76.8 (0.3)	80.2 (0.2)	81.9 (0.2)	83.0 (0.3)	83.5 (0.3)	84.3 (0.4)	84.2 (0.4)	85.7 (0.5)
	PPI	76.7 (0.5)	80.1 (0.3)	81.8 (0.3)	82.8 (0.3)	83.4 (0.3)	84.2 (0.2)	84.1 (0.4)	85.5 (0.6)
	CoExp	77.0 (0.4)	80.2 (0.3)	81.9 (0.2)	83.0 (0.4)	83.5 (0.5)	84.3 (0.1)	84.2 (0.5)	85.7 (0.7)
	All	**77.8 (0.7)**	**81.1 (0.2)**	**82.7 (0.3)**	**83.7 (0.2)**	**84.5 (0.3)**	**84.8 (1.3)**	**85.0 (0.3)**	**86.0 (0.4)**
TCGA-5000	Basic	76.1 (0.6)	79.2 (0.5)	80.6 (0.4)	81.5 (0.4)	82.1 (0.6)	82.2 (0.4)	82.5 (0.4)	83.4 (0.6)
	GGI	77.9 (0.4)	81.4 (0.3)	82.8 (0.5)	84.1 (0.2)	84.9 (0.4)	85.3 (0.3)	85.1 (0.5)	86.1 (0.5)
	PPI	77.9 (0.4)	81.1 (0.3)	82.9 (0.4)	84.0 (0.3)	84.8 (0.4)	85.2 (0.4)	85.2 (0.3)	86.1 (0.5)
	CoExp	78.1 (0.4)	81.5 (0.3)	82.9 (0.4)	84.0 (0.2)	84.9 (0.4)	85.3 (0.3)	85.2 (0.4)	86.2 (0.4)
	All	**78.6 (0.6)**	**81.5 (0.4)**	**83.5 (0.3)**	**84.6 (0.2)**	**85.6 (0.4)**	**85.9 (0.2)**	**85.7 (0.4)**	**86.6 (0.4)**

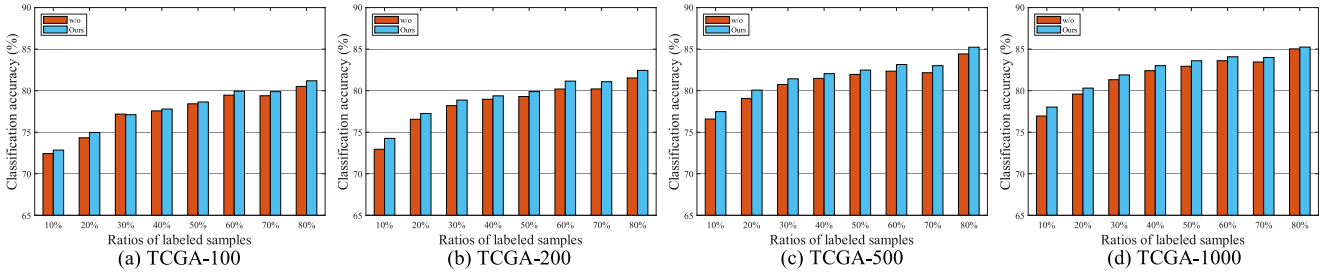

Fig. 2. Classification accuracy comparison between MPK-GNN and its variant without contrastive learning under different labeling ratios.

MPK-GNN model with a variant that lacks contrastive learning loss. Specifically, this variant discards the projection module, and each input prior knowledge graph has its own learning network. Fig. 2 depicts the performance comparison between MPK-GNN and this variant, which is denoted as w/o in the experiment. The experimental results show that the used contrastive learning framework can enhance the performance of the learning algorithm in most cases. It demonstrates that contrastive learning can be utilized to better embed the information contained in multiple prior knowledge graphs into the learned embedding by maximizing the agreement among prior networks. Compared to the w/o variant, the designed MPK-GNN model has fewer network parameters because the contrastive learning framework is shared by all prior knowledge graphs.

Predicted Label Distribution Analysis: We conduct an additional experiment to examine misclassification across all categories by analyzing the predicted label distribution. This study is conducted on the TCGA-1000 dataset with 80% training samples and 10% samples used for testing. Based on the test set, we use the true class and predicted class to generate a confusion matrix, which is standardized for each individual class. As shown in Fig. 3, the top five most miss-classification categories highlighted by dark blue boxes including C7: Mixed (Chr 9 del), C1: STAD (EBV-CIMP), C17: BRCA Chr 8q amp, C21: DLBC and C13: Mixed (Chr 8 del). Two of them are mixed molecular subtypes, which contain more than one traditional tissue of origin cancer type. The observation demonstrates that it is challenging to predict mixed cancer classes rather than other subtypes.

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

Predicted Label

Ground-truth	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24	C25	C26	C27	C28	SUM
C1	0.500	0	0	0	0	0	0	0	0	0	0	0	0	0	0.167	0	0	0.167	0	0.167	0	0	0	0	0	0	0	0	6
C2	0	0.667	0	0.111	0	0	0	0	0	0.111	0	0	0	0	0	0	0	0	0.111	0	0	0	0	0	0	0	0	0	9
C3	0	0	0.818	0	0.023	0	0.023	0	0	0.023	0	0.023	0.023	0	0.068	0	0	0	0	0	0	0	0	0	0	0	0	0	44
C4	0	0	0	0.976	0	0	0	0	0	0	0	0	0	0	0	0	0	0.024	0	0	0	0	0	0	0	0	0	0	42
C5	0	0	0.107	0	0.821	0	0	0	0.036	0	0	0	0	0	0	0	0	0	0	0	0	0.036	0	0	0	0	0	0	28
C6	0	0	0	0	0	0.953	0	0.023	0	0	0	0	0.023	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	43
C7	0	0	0.056	0	0	0	0.278	0	0	0.167	0	0	0	0	0.222	0	0.056	0	0	0.056	0	0	0	0	0.056	0.056	0.056	0	18
C8	0	0	0	0	0	0	0	0.917	0	0.021	0	0	0	0	0	0	0	0	0	0.042	0	0	0	0	0	0	0.021	0	48
C9	0	0	0	0	0	0	0	0	0.867	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.133	15
C10	0	0	0	0	0	0.097	0	0	0	0.629	0	0	0.032	0.016	0	0	0.065	0.016	0	0.032	0	0	0	0	0	0	0.113	0	62
C11	0	0	0	0	0	0	0	0	0	0	0.976	0	0	0	0	0	0	0	0	0	0	0	0.024	0	0	0	0	0	41
C12	0	0	0	0	0	0	0	0	0	0	0	1.000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	44
C13	0	0	0	0	0	0.067	0.033	0	0	0.067	0	0	0.567	0.033	0	0	0.100	0	0.033	0.033	0	0.033	0	0	0	0	0.033	0	30
C14	0	0	0.024	0	0	0	0	0	0	0	0	0	0	0.829	0	0	0	0	0	0.122	0.024	0	0	0	0	0	0	0	41
C15	0	0	0	0	0	0	0.038	0	0	0	0	0	0	0	0.962	0	0	0	0	0	0	0	0	0	0	0	0	0	26
C16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.000	0	0	0	0	0	0	0	0	0	0	0	0	46
C17	0	0.071	0	0	0	0	0	0	0	0.071	0	0	0	0	0	0	0.500	0	0.143	0	0	0	0	0	0.214	0	0	0	14
C18	0	0	0	0.121	0	0	0	0	0	0	0	0	0	0	0	0	0	0.758	0	0.061	0	0	0	0	0	0	0.061	0	33
C19	0	0.017	0	0	0	0	0	0	0	0	0	0	0.017	0	0	0	0.017	0	0.915	0.017	0	0	0	0	0.017	0	0	0	59
C20	0	0	0	0.013	0	0	0.013	0.013	0	0.039	0	0	0	0.066	0	0	0	0	0.013	0.803	0	0	0	0	0	0	0.039	0	76
C21	0	0	0.500	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.500	0	0	0	0	0	0	0	2
C22	0	0	0.048	0	0	0	0	0.095	0	0	0	0	0	0.048	0	0	0	0	0	0	0	0.810	0	0	0	0	0	0	21
C23	0	0	0	0	0.053	0	0	0	0	0	0	0	0.053	0	0	0	0	0	0	0	0	0	0.895	0	0	0	0	0	19
C24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.000	0	0	0	0	13
C25	0	0	0	0	0	0	0	0	0	0.042	0	0	0	0	0	0	0	0	0.042	0	0	0	0	0	0.917	0	0	0	24
C26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.000	0	0	37
C27	0	0	0	0	0	0.019	0.019	0	0	0.077	0	0	0	0	0	0	0	0.019	0	0.019	0	0	0	0	0.019	0	0.827	0	52
C28	0	0	0	0	0	0	0	0	0.014	0	0	0	0	0	0	0	0	0	0	0.014	0	0	0	0	0	0	0	0.973	74

Fig. 3. Row-normalized confusion matrix derived from the test set of TCGA-1000 with 80% training samples.

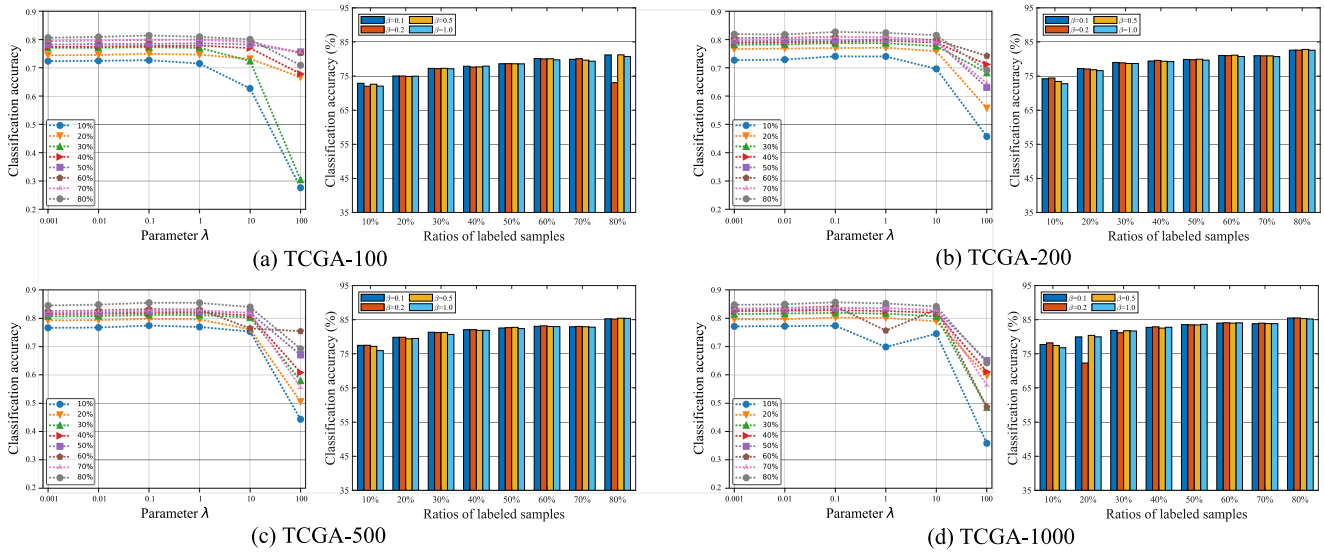

Fig. 4. Sensitivity analysis of parameters λ and β for MPK-GNN measured by classification accuracy under different proportions of labels.

Sensitivity of Hyper-parameters λ and β: To verify the sensitivity of classification performance about the weight of contrastive learning loss λ and the temperature parameter β, we run the proposed MPK-GNN framework with different combinations on four experimental datasets, including TCGA-100, TCGA-200, TCGA-500 and TCGA-1000. In detail, the values of λ are taken from 0.001 to 100 and the hyper-parameter β ranges in {0.1, 0.2, 0.5, 1.0}. We fix the value of one hyper-parameter when analyzing the sensitivity of another hyper-parameter. The results are depicted in Fig. 4 for an intuitive observation. We can find that the proposed MPK-GNN framework has a stable performance when varying the values of both λ and β in most situations. This characteristic is important for deep learning-based multi-omics analysis algorithms because they are vulnerable to the values of hyper-parameters. In this respect, the robustness of MPK-GNN makes it more suitable to be extended for different multi-omics data analysis tasks.

Statistical test of classification performance: In addition, we apply the Bonferroni-Dunn post-hoc test to verify the significant differences among all algorithms on the cancer molecular subtype classification task. Specifically, there is a significant difference in performance between algorithms if the distance between their average rankings exceeds the Critical Difference (CD). The CD value is computed by $\mathrm{CD} = q_\alpha \sqrt{a(a+1)/6\,N}$, where q_α is the critical value, a represents the number of compared algorithms and N denotes the number of datasets. In detail,

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.

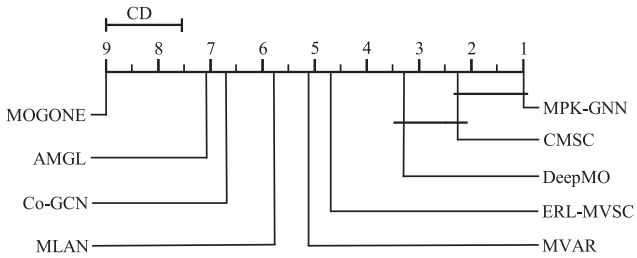

Fig. 5. Statistical comparison of classification accuracy of different learning algorithms using the Bonferroni-Dunn test.

we compare all algorithms except for single-view methods in this test, with a total of nine algorithms. The results of different datasets with various labeling ratios are all used for comparison, which can be seen as 48 datasets. According to Table 5 in [30], $\mathrm{CD} = 1.3964$ when $\alpha = 0.1$. The comparison result is shown in Fig. 5, where the scale value on the horizontal axis represents the average ranking. We can observe that the performance of the proposed algorithm is significantly superior to all compared algorithms except for CMSC from a statistical perspective.

V. Conclusion

In this article, we proposed a GNN-based deep learning framework termed MPK-GNN for multi-omics data analysis. The designed shared contrastive learning architecture efficiently captures helpful information from multiple prior biological knowledge to improve the performance of the computational model, especially when there are a few features or training samples available. Based on the multi-level representation learning module, the proposed MPK-GNN framework can be flexibly extended to other multi-omics analysis tasks by implementing the task-specific with suitable networks. Comprehensive experiments demonstrate that the proposed framework has achieved competitive results on seven benchmark datasets concurrently. Our future works include refining the sample-based module to capture more information from the input multiple features and verifying the superiority of MPK-GNN on more multi-omics data analysis tasks.

References

[1] J. S. Reis-Filho and L. Pusztai, "Gene expression profiling in breast cancer: Classification, prognostication, and prediction," *Lancet*, vol. 378, no. 9805, pp. 1812–1823, 2011.

[2] Y. Hasin, M. Seldin, and A. Lusis, "Multi-omics approaches to disease," *Genome Biol.*, vol. 18, no. 1, pp. 1–15, 2017.

[3] H. Sharifi-Noghabi, O. Zolotareva, C. C. Collins, and M. Ester, "MOLI: Multi-omics late integration with deep neural networks for drug response prediction," *Bioinformatics*, vol. 35, no. 14, pp. i501–i509, 2019.

[4] C. Dimitrakopoulos et al., "Network-based integration of multi-omics data for prioritizing cancer genes," *Bioinformatics*, vol. 34, no. 14, pp. 2441–2448, 2018.

[5] B. Xie, Z. Yuan, Y. Yang, Z. Sun, S. Zhou, and X. Fang, "MOBCdb: A comprehensive database integrating multi-omics data on breast cancer for precision medicine," *Breast Cancer Res. Treat.*, vol. 169, no. 3, pp. 625–632, 2018.

[6] I. Lauriola, A. Lavelli, and F. Aiolli, "An introduction to deep learning in natural language processing: Models, techniques, and tools," *Neurocomputing*, vol. 470, pp. 443–456, 2022.

[7] Y. Pan et al., "Compressing recurrent neural networks with tensor ring for action recognition," in *Proc. 33rd AAAI Conf. Artif. Intell.*, 2019, pp. 4683–4690.

[8] K. Chaudhary, O. B. Poirion, L. Lu, and L. X. Garmire, "Deep learning-based multi-omics integration robustly predicts survival in liver cancer," *Clin. Cancer Res.*, vol. 24, no. 6, pp. 1248–1259, 2018.

[9] S. Albaradei, F. Napolitano, M. A. Thafar, T. Gojobori, M. Essack, and X. Gao, "Metacancer: A deep learning-based pan-cancer metastasis prediction model developed using multi-omics data," *Comput. Struct. Biotechnol. J.*, vol. 19, pp. 4404–4411, 2021.

[10] X. Zhang, Y. Xing, K. Sun, and Y. Guo, "OmiEmbed: A unified multi-task deep learning framework for multi-omics data," *Cancers*, vol. 13, no. 12, 2021, Art. no. 3047.

[11] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learn. Representations*, 2017, pp. 1–14,.

[12] P. Veličć, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. 6th Int. Conf. Learn. Representations*, 2018, pp. 1–12,.

[13] R. Feng, Y. Xie, M. Lai, D. Z. Chen, J. Cao, and J. Wu, "AGMI: Attention-guided multi-omics integration for drug response prediction with graph neural networks," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2021, pp. 1295–1298.

[14] T. Wang et al., "Mogonet integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification," *Nature Commun.*, vol. 12, no. 1, pp. 1–13, 2021.

[15] B. Li, T. Wang, and S. Nabavi, "Cancer molecular subtype classification by graph convolutional networks on multi-omics data," in *Proc. 12th ACM Conf. Bioinf., Comput. Biol., & Health Inform.*, 2021, pp. 1–9.

[16] M. Q. Ding, L. Chen, G. F. Cooper, J. D. Young, and X. Lu, "Precision oncology beyond targeted therapy: Combining omics data with machine learning matches the majority of cancer cells to effective therapeutic-sassigning cancers to effective drugs with Big Data," *Mol. Cancer Res.*, vol. 16, no. 2, pp. 269–278, 2018.

[17] Y. Lin, W. Zhang, H. Cao, G. Li, and W. Du, "Classifying breast cancer subtypes using deep neural networks based on multi-omics data," *Genes*, vol. 11, no. 8, 2020, Art. no. 888.

[18] K. Jha, S. Saha, and H. Singh, "Prediction of protein–protein interaction using graph neural networks," *Sci. Rep.*, vol. 12, no. 1, pp. 1–12, 2022.

[19] T. Nguyen, G. T. Nguyen, T. Nguyen, and D.-H. Le, "Graph convolutional networks for drug response prediction," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 19, no. 1, pp. 146–154, Jan./Feb. 2022.

[20] Q. Liu, Z. Hu, R. Jiang, and M. Zhou, "DeepCDR: A hybrid graph convolutional network for predicting cancer drug response," *Bioinf.*, vol. 36, no. Supplement_2, pp. i911–i918, 2020.

[21] W. Peng, T. Chen, and W. Dai, "Predicting drug response based on multi-omics fusion and graph convolution," *IEEE J. Biomed. Health Inform.*, vol. 26, no. 3, pp. 1384–1393, Mar. 2022.

[22] C. Yin et al., "Molecular subtyping of cancer based on robust graph neural network and multi-omics data integration," *Front. Genet.*, vol. 13, 2022, Art. no. 884028.

[23] S. Li, W.-T. Li, and W. Wang, "Co-GCN for multi-view semi-supervised learning," in *Proc. 34th AAAI Conf. Artif. Intell.*, 2020, pp. 4691–4698.

[24] A. Huang, Z. Wang, Y. Zheng, T. Zhao, and C.-W. Lin, "Embedding regularizer learning for multi-view semi-supervised classification," *IEEE Trans. Image Process.*, vol. 30, pp. 6997–7011, 2021.

[25] N. D. Nguyen and D. Wang, "Multiview learning for understanding functional multiomics," *PLoS Comput. Biol.*, vol. 16, no. 4, 2020, Art. no. e1007677.

[26] R. Oughtred et al., "The biogrid database: A comprehensive biomedical resource of curated protein, genetic, and chemical interactions," *Protein Sci.*, vol. 30, no. 1, pp. 187–200, 2021.

[27] D. Szklarczyk et al., "STRING v11: Protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets," *Nucleic Acids Res.*, vol. 47, no. D1, pp. D607–D613, 2019.

[28] P. Khosla et al., "Supervised contrastive learning," in *Proc. 34th Annu. Conf. Neural Inf. Process. Syst.*, 2020, pp. 18661–18673.

[29] M. J. Goldman et al., "Visualizing and interpreting cancer genomics data via the Xena platform," *Nature Biotechnol.*, vol. 38, no. 6, pp. 675–678, 2020.

[30] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.

Authorized licensed use limited to: Politecnico di Milano. Downloaded on February 23,2026 at 10:03:29 UTC from IEEE Xplore. Restrictions apply.